

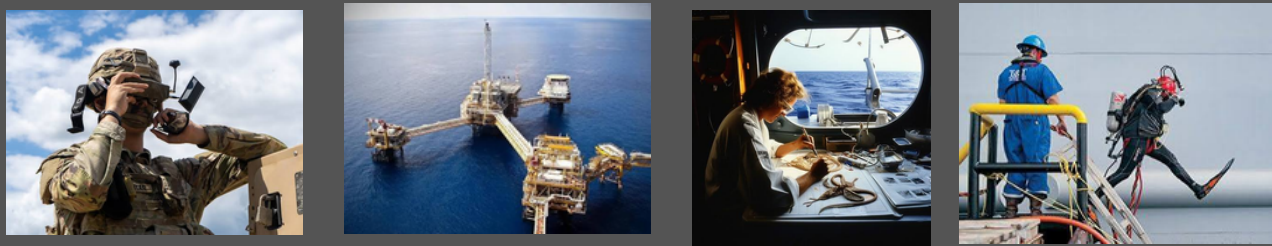
Day 2 - Literature Review - Domain + ML

UnderWater Communication :

Underwater communication is the transmission of information between underwater devices using sound, radio waves, or light.

Why is it Needed :

- Defense & Naval Operations: Communication between submarines, ships, and divers.
- Marine Research: Ocean monitoring, climate studies, and marine life observation.
- Offshore Industry: Pipeline inspection and underwater equipment monitoring.
- Rescue Operations: Diver-to-diver and diver-to-surface communication.

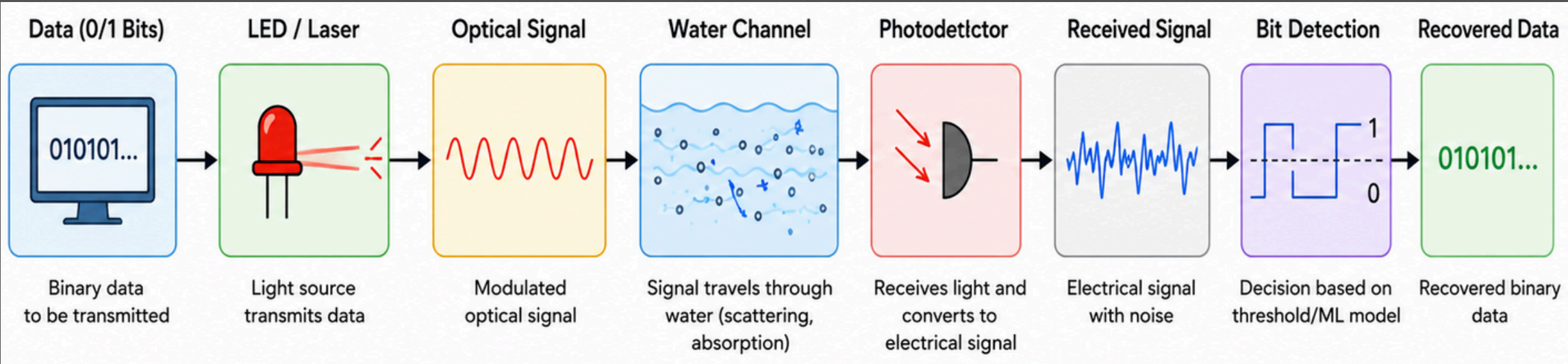


Communication Techniques :

Technique	Range	Bandwidth
Acoustic	Long (km)	Low (kbps)
Radio Frequency	Short	Moderate
Optical (UWOC)	Short	High (Gbps)



UWOC Working



What is Fixed Threshold ?

A Fixed Threshold is a simple method used to detect transmitted bits based on the received signal strength.

Received Signal > Threshold → Bit 1
Received Signal ≤ Threshold → Bit 0

Received Signal	Predicted Bit
0.8	1
0.7	1
0.2	0
0.3	0

Threshold = 0.5

How Turbidity Affects It ?

Turbidity (cloudiness of water) causes absorption and scattering of light.

As turbidity increases:

- Received signal becomes weaker.
- Signal values of bit 0 and 1 start overlapping.
- Fixed threshold can no longer separate them correctly.

Threshold = 0.5

Actual Bit	Received Signal	Predicted
0	0.55	1 ❌
1	0.48	0 ❌

Drawback :

- Same threshold is used for all water conditions.
- Does not adapt to different NTU levels.
- Leads to higher Bit Error Rate (BER).

Machine Learning :

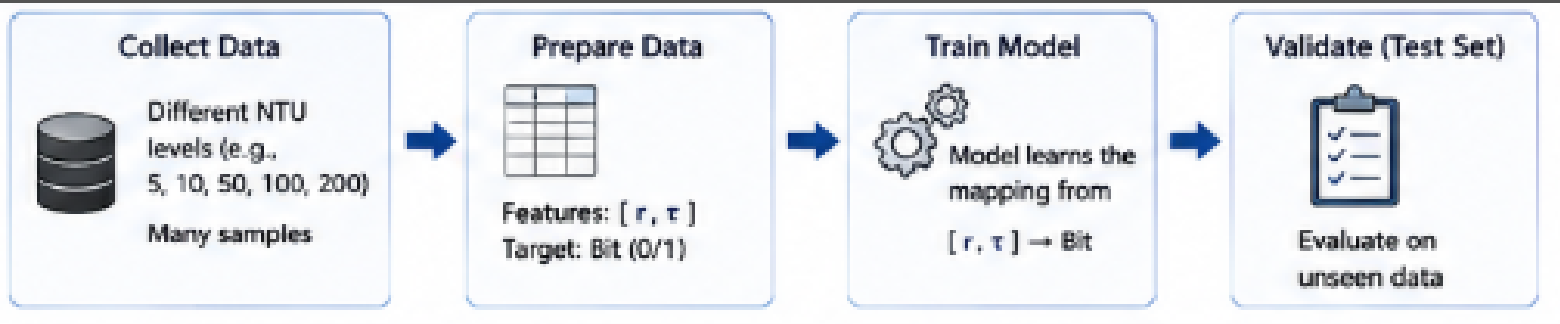
The idea is to let the ML model learn the relationship between the received optical signal, turbidity (NTU), and the actual transmitted bit instead of using one fixed threshold

ML Type : Supervised Machine Learning (Binary Classification)

- Goal: Classify the transmitted bit as 0 or 1.
- Input: Received Signal + NTU
- Output: Predicted Bit (0/1)

Working :

- Multiple Inputs: Uses received signal and NTU/turbidity.
- Learns Patterns: Learns from labelled 0 and 1 bits.
- Handles Turbidity: Learns signal patterns across different NTU levels.
- Reduces Errors: Improves bit detection and reduces BER.
- Compares Models: Compares ML models with fixed threshold using accuracy, F1, BER, and inference time.



Reviewed Methods :

Method	Approach	Advantages	Limitations
Fixed Threshold	Rule-based	Simple, Fast	Fails under varying turbidity
Logistic Regression	Linear Classifier	Interpretable	Limited for complex patterns
SVM	Margin-based Classification	Good generalization	Higher computation
Random Forest	Ensemble Trees	Handles non-linearity	Larger model size
MLP (Optional)	Neural Network	Learns complex patterns	Requires more training data

Evaluation Metrics :

- Accuracy: Overall correct predictions.
- Precision: Correctly predicted positive bits.
- Recall: Detected actual positive bits.
- F1-Score: Balance of precision and recall.
- Confusion Matrix: Shows classification errors.
- BER: Percentage of incorrectly detected bits.
- Inference Time: Time taken for prediction.

Success Criteria :

- High Accuracy: Correctly classify most bits.
- Low BER: Minimize bit detection errors.
- Robust Across NTU: Consistent performance at different turbidity levels.
- Fast Inference: Quick prediction with low computational cost.